

Assignment 2

BSSE23-A/B

Multivariable Calculus | (MT101T) | Fall '25

Name: _____ Reg. No.: _____

Total Points = 75

Q1. [CLO-2] [10pts]

The *body mass index* (BMI) of a person is defined by $B(m, h) = \frac{m}{h^2}$ where m is the person's mass (in kilograms) and h is the height (in meters).

- Draw the level curves

$$B(m, h) = k = \{18.5, 25, 30, 40\}$$

A rough guideline is that a person is underweight if the BMI is less than 18.5; optimal if the BMI lies between 18.5 and 25; overweight if the BMI lies between 25 and 30; and obese if the BMI exceeds 30.

- Shade the region corresponding to optimal BMI.
- A person who weighs 62 kg and is 1.52 m tall fall into which category? Draw the point on the contour plot.

Q2. [CLO-2] [15pts]

Find the indicated partial derivatives

- | | |
|-----------------------------|---|
| a. $W = \sqrt{u + v^2}$ | $\frac{\partial^3 W}{\partial u^2 \partial v}$ |
| b. $V = \ln(r + s^2 + t^3)$ | $\frac{\partial^3 V}{\partial r \partial s \partial t}$ |
| c. $w = \frac{x}{y+2z}$ | $\frac{\partial^3 w}{\partial z \partial y \partial x}, \frac{\partial^3 w}{\partial x^2 \partial y}$ |

Q3. [CLO-2] [15pts]

Use the Chain Rule to find the indicated partial derivatives.

- a. $T = \frac{v}{2u+v}$ where $u = pq\sqrt{r}$ $v = p\sqrt{qr}$
 $\frac{\partial T}{\partial p}, \frac{\partial T}{\partial q}, \frac{\partial T}{\partial r}$ when $p = 2, q = 1, r = 4$
- b. $w = xy + yz + zx$ where $x = r\cos\theta, y = r\sin\theta, z = r\theta$
 $\frac{\partial w}{\partial r}, \frac{\partial w}{\partial \theta}$, when $r = 2, \theta = \frac{\pi}{2}$
- c. $P = \sqrt{u^2 + v^2 + w^2}$ where $u = xe^y, v = ye^x, w = e^{xy}$
 $\frac{\partial P}{\partial x}, \frac{\partial P}{\partial y}$ when $x = 0, y = 2$

Q4. [CLO-2] [10pts]

- a. Given the function $f(x, y) = 4x - 2y + 3$, sketch the level curve that passes through $P(1, 2)$, and draw the gradient vector at P .

- b. Given the function $f(x, y) = \frac{y}{x^2}$, sketch the level curve that passes through $P(-2, 2)$, and draw the gradient vector at P .

Q5. [CLO-2] [10pts]

- a. Find $D_{\mathbf{u}}f$ at point $P(0, 2, 3)$, where $f(x, y, z) = ye^{xz} + z^2$ in the direction of $\mathbf{u} = \frac{2}{7}\hat{i} - \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k}$
- b. Find the directional derivative of $f(x, y, z) = y - \sqrt{x^2 + z^2}$ at $P(-3, 1, 4)$ in the direction of $\mathbf{a} = 2\mathbf{i} - 2\mathbf{j} - \mathbf{k}$

Q6. [CLO-2] [15pts]

You are given a surface $x^2 + y^2 + 4z^2 = 12$.

- a. Find an equation of the tangent plane to the surface at the point $(2, 2, 1)$.
- b. Find parametric equations of the line that is normal to the surface at the point $(2, 2, 1)$.
- c. Find the acute angle that the tangent plane at the point $(2, 2, 1)$ makes with the xy -plane.